

VIVEK KUMAR SINGH

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience designing scalable enterprise applications across frontend and AI domains. Extensive expertise in Angular, TypeScript, frontend architecture, performance optimization, and reusable component design, combined with hands-on experience building production-ready Generative AI solutions using Python, FastAPI, LangChain, OpenAI APIs, Retrieval-Augmented Generation (RAG), semantic search, and conversational AI. Proven ability to design maintainable architectures, collaborate across cross-functional teams, and deliver scalable software from user interface to AI backend services.

Core Skills

- **AI & GenAI Development (Learning & Building):** Python, FastAPI, Generative AI, LLM API Integration (OpenAI, Gemini), LangChain, Prompt Engineering, Embeddings, Vector Search, FAISS,
- **Frontend:** Angular (2–16), TypeScript, JavaScript, HTML5, CSS3, Bootstrap
- **Frontend Architecture:** Component Architecture, RxJS, Angular Signals, Lazy Loading, Performance Optimization
- **Visualization & UI Libraries:** D3.js, Kendo UI, Phaser2, Phaser3, PixiJS
- **Tools & Platform:** Git, GitHub, Azure Devops, Agile/Scrum
- **Development Practices:** Agile & Scrum Methodologies, Clean & Scalable Code Practices, TDD CI/CD, UI/UX, Data Structure and Algorithms, System Design, Object Oriented Programming(OPPs)

WORK EXPERIENCE

Associate Consultant | Gen AI Developer

Tata Consultancy Services(TCS) | Sept 2025 – Present

- Designed and implemented an enterprise **Retrieval-Augmented Generation (RAG)** architecture supporting document ingestion, semantic chunking, embedding pipelines, vector retrieval, metadata-aware search, and source citations.
- Designed a pluggable **LLM integration** framework with adapter-based support for OpenAI, Azure OpenAI, Anthropic, and Gemini, enabling streaming responses and simplifying future provider integrations.
- Built a scalable conversation platform with JWT-secured authentication, persistent conversation lifecycle management, streaming responses, and PostgreSQL-backed message history optimized for enterprise AI workloads.
- Built reusable prompt and conversation context pipelines using **LangChain**, enabling consistent **prompt generation**, **context retrieval**, and **chat memory management**.
- Implemented content-**hash-based deduplication** and metadata lifecycle management to improve retrieval consistency and prevent redundant vector indexing within the RAG pipeline.
- Maintain and enhance enterprise healthcare web applications built with Angular and TypeScript, ensuring reliability and performance in production environments.
- Identified performance bottlenecks and implemented optimizations (lazy loading, change detection), improving page load time by 20–25%.
- **Technologies:** Angular 14, TypeScript, JavaScript, FastAPI, OpenAI APIs, LangChain, FAISS, Git, GitHub, Azure, HTML, CSS, Bootstrap, Agile.

Senior Associate | Lead UI Developer PwC

India | Aug 2021 – Aug 2025

- Design and develop visually appealing and responsive UI components, focusing on functionality, aesthetics, and usability, with a primary focus on Angular frameworks.
- Designed and developed scalable Angular-based UI systems and reusable components, improving development efficiency by 35%.
- Created finance dashboards that enables administrators to visualize the current financial status through interactive charts(using **d3**) and graphs enhancing decision-making capabilities.
- Implemented **state management** solutions using **RxJS**, leveraging **observables**, **Angular Signals** to efficiently manage application state and ensure seamless data flow in Angular applications.

- Collaborate with business analysts to plan, organize, and prioritize UI tasks, establishing a well-structured development road-map.
- Successfully upgraded several applications from earlier Angular versions to Angular 14, Angular 16 enhancing performance, security, and user experience.
- Created **custom pipes** and **libraries**, leveraged **web workers** and implemented **lazy loading techniques** to optimize application performance.
- Technologies: - Angular 16, JavaScript , Kendo, Git, GitHub, Azure, HTML, CSS, Bootstrap, Agile, Typescript

Game Developer | Game Owner

Ingenuity Gaming | Jan 2020 – Jul 2021

- Took complete ownership of the game development process, collaborating with game artists to integrate design elements and assets, and working with mathematics teams to implement game rules.
- Developed game components and features utilizing object-oriented programming (OOP) principles to ensure scalability and maintainability of the code base.
- Identified UI performance bottlenecks and implemented optimizations, including preload and post-load techniques, to enhance gameplay fluidity and reduce loading times upto 30%.
- Worked closely with product owners and managers to define and implement a structured road-map, ensuring efficient game development.
- Developed and launched multiple production slot games including All Sports, All About Christmas, Book of Inti, and 5 Fruit Frenzy, used across online gaming platforms.
- Technologies: - JavaScript, Typescript, Phaser2, Phaser3, pixi.js , Git, GitHub, Azure, HTML5, Agile

Associate Engineer | UI Developer

Tech Mahindra Ltd | Feb 2017 – Dec 2019

- Requirement understanding, application design, development and unit testing.
- Coordinate with the Product Owner and Development team.
- Facilitate different agile ceremonies encouraging and demanding self-organization from the development team, unit testing, code reviews and technical help to the team.
- Keeping Application Lifecycle Management (ALM) updated and following scrum practices, ensuring all Sprint activities are time boxed.
- Technologies: Java, Angular2, Bootstrap, Scrum, Agile Methodology.

WORK SAMPLE(S):

- **Project: RoleFit | 2026 (In Progress)**: AI-powered resume tailoring platform built with Angular 18 (frontend) and Python FastAPI (backend), integrating OpenAI and Gemini APIs for intelligent resume analysis and job-fit scoring.
- **Project: Giggly UI| Jan 2025**: Developed Giggly UI, a modern Angular-based UI component library designed to provide a playful yet professional user experience. It includes customization, responsive components such as buttons, drop-downs, and form elements, ensuring seamless integration with Angular forms. Deployed the project using Firebase for accessibility and scalability. *Link:* <https://mygigglyui.web.app/>.

CERTIFICATIONS

- Gen AI & Agentic AI for Developers- IIT Patna(Feb 2026 - Ongoing)
- UI Design Specialization - California School of Arts (Aug 2023)

EDUCATION

- Executive MBA(2026) | Indian Institute of Management Amritsar(IIM)
- Bachelor of Computer Science and Engineering (2016) | Visvesvaraya Technological University

CO-CURRICULAR ACTIVITIES/ACHIEVEMENT

- All India ranked 328 in Coding-Ninjas hackathon.
- Winner of KVS state-level chess tournament.
- Represented college at the state level in chess for three years.